Observation of the Integer Quantum Hall Effect

in a GaAs/AlGaAs Two-Dimensional Electron Gas

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Statement on the Use of Generative Artificial Intelligence

Generative artificial intelligence (OpenAI ChatGPT) was used at several points during this project as a support tool. It was used to suggest example code snippets during the development of data acquisition software for lock-in amplifiers and a Keithley source meter. It also assisted in checking understanding of physics and mathematical concepts, as well as with rephrasing text, LaTeX formatting, and spelling and grammar checks.

All physics-related decisions, experimental work, code implementation, data analysis, figures, interpretation of results, and final conclusions were carried out by the author.

Abstract

Magnetotransport measurements were performed on a GaAs/AlGaAs two-dimensional electron gas to investigate the integer quantum Hall effect under laboratory-scale cryogenic conditions. The primary aim of this work was to demonstrate the defining transport signatures of the effect and to extract key electronic parameters using internally consistent classical and quantum analysis methods. Measurements of the Hall resistance R_{xy} and longitudinal resistance R_{xx} were carried out as functions of perpendicular magnetic field at temperatures between 3 K and 5 K. At low magnetic fields, the Hall resistance exhibits the expected linear behaviour of the classical Hall effect, while at higher fields a well-defined Hall resistance plateau is observed at filling factor $\nu = 2$, centred at $B_{\nu=2} = 3.23\,\mathrm{T}$, accompanied by strong suppression of the longitudinal resistance. Within the plateau window, the measured Hall resistance deviates from the ideal quantised value $h/2e^2$ by only tens of parts per million. The carrier density was extracted using four independent methods, yielding a consistent value of $n_s = 1.57 \times 10^{15}\,\mathrm{m}^{-2}$ with a maximum fractional spread of approximately 0.3 %. From low-field longitudinal transport, a mobility of $\mu \approx 9.5 \times 10^5\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ and a corresponding transport lifetime of $\tau_{\mathrm{tr}} \approx 3.5 \times 10^{-11}\,\mathrm{s}$ were obtained. The temperature dependence of the Shubnikov–de Haas oscillations is consistent with thermal broadening of Landau levels, while the invariance of the low-field Hall slope confirms that the carrier density remains constant over the measured temperature range. Despite experimental constraints on temperature stability and geometric uncertainty, the essential signatures of the integer quantum Hall effect are robustly observed. This work demonstrates that a laboratory-scale cryogenic apparatus is sufficient to access the quantum Hall regime and to perform a quantitatively consistent analysis of two-dimensional electron transport.

Contents

1 Introduction

1.1 Motivation and Context

Low-dimensional electronic systems exhibit transport phenomena that have no analogue in bulk materials. Among the most striking of these is the integer quantum Hall effect, in which the Hall resistance becomes quantised at integer multiples of h/e^2 when a two-dimensional electron gas is subjected to a strong perpendicular magnetic field at low temperature [1].

The quantum Hall effect is remarkable not only for its conceptual significance, but also for its exceptional robustness. The quantised Hall resistance is insensitive to microscopic details such as disorder and sample geometry, making it one of the most precise and reproducible phenomena in condensed matter physics [1, 2]. This robustness underpins its use as a resistance standard [1] and motivates its continued study in both fundamental and applied contexts.

From an experimental perspective, the quantum Hall effect provides a powerful demonstration of how quantum mechanics, dimensionality, and disorder combine to produce macroscopic transport behaviour. Observing its defining signatures in a laboratory setting therefore serves as a stringent test of experimental control and analysis techniques in low-temperature condensed matter physics.

1.2 The Two-Dimensional Electron Gas

The quantum Hall effect is most readily observed in a two-dimensional electron gas (2DEG), where electronic motion is confined to a plane. In this work, the 2DEG is realised in a GaAs/AlGaAs heterostructure, in which band bending at the interface creates a narrow potential well that confines electrons in the growth direction [3].

This confinement quantises motion perpendicular to the interface, while allowing free motion in the plane, producing an effectively two-dimensional electronic system. Such heterostructures can achieve high carrier mobilities due to spatial separation of charge carriers from ionised donors, making them well suited for the observation of quantum transport phenomena [3]. The formation of the two-dimensional electron gas at a GaAs/AlGaAs heterointerface is illustrated schematically in Fig. 1.

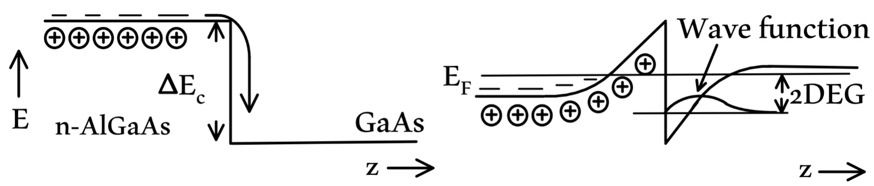

Figure 1: Schematic illustration of the formation of a two-dimensional electron gas at a GaAs/AlGaAs heterointerface. Band bending creates a triangular quantum well that confines electrons in the growth direction, producing an effectively two-dimensional electronic system. Adapted from [4].

The key transport parameters characterising the 2DEG include the carrier density, mobility, and scattering time. These parameters determine the magnetic-field and temperature scales at which Landau quantisation, Shubnikov–de Haas oscillations, and quantum Hall plateaux become experimentally accessible [3].

1.3 From Classical to Quantum Hall Transport

At low magnetic fields, transverse transport (the Hall response measured perpendicular to the current direction) in a conducting system is well described by the classical Hall effect, in which the Hall resistance increases linearly with magnetic field. This regime provides a straightforward means of extracting the carrier density from the Hall slope.

At higher magnetic fields and sufficiently low temperatures, however, this classical description breaks down. Landau quantisation of cyclotron motion leads to discrete energy levels, and the electronic density of states becomes strongly modulated. In the presence of disorder, this gives rise to the localisation physics that underpins the integer quantum Hall effect [3].

Experimentally, the transition from classical to quantum Hall transport is marked by the appearance of Shubnikov–de Haas oscillations in the longitudinal resistance, associated with longitudinal transport (current and voltage measured parallel to the applied electric field), followed by the emergence of quantised Hall plateaux accompanied by suppression of longitudinal dissipation [5]. Understanding this progression provides a coherent framework for interpreting magnetotransport measurements in two-dimensional systems.

1.4 Aims of This Work

The primary aim of this project is to demonstrate the defining transport signatures of the integer quantum Hall effect in a GaAs/AlGaAs two-dimensional electron gas using a laboratory-scale cryogenic apparatus. Successful observation is defined by the appearance of a Hall resistance plateau at an integer filling factor accompanied by strong suppression of the longitudinal resistance.

A secondary aim is to extract key electronic transport parameters, including carrier density and mobility, using multiple independent classical and quantum methods, and to assess the internal consistency of these estimates. Agreement between parameters extracted from distinct measurement regimes is used as a criterion for the robustness of the analysis.

1.5 Structure of the Report

The remainder of this report is organised as follows. Section 2 introduces the theoretical framework underlying the classical and quantum Hall effects, including Landau quantisation, localisation, and edge-state transport. This section also discusses the relevant energy scales and motivates the experimental requirement for low temperatures and high magnetic fields.

The experimental apparatus and measurement techniques are described in Sec. 3. Section 4 presents the experimental results, including magnetotransport measurements, carrier-density and mobility extraction, and analysis of Hall quantisation and temperature dependence. These results are interpreted and compared to established literature in Sec. 5, before the main conclusions are summarised in Sec. 6.

2 Theory: From Classical Hall to Quantum Hall

2.1 Classical Hall Effect

The classical Hall effect provides a useful baseline for understanding transverse transport in a conductor subjected to a perpendicular magnetic field [6]. When a current flows through a conducting channel, the Lorentz force deflects charge carriers, leading to the development of a transverse Hall voltage.

In the classical regime, the Hall resistance is given by

$$R_{xy} = \frac{B}{n_s e} \tag{1}$$

where B is the magnetic field, n_s is the carrier density, and e is the elementary charge. At sufficiently low magnetic fields, this relation predicts a linear dependence of R_{xy} on B, with the slope directly yielding the carrier density [6].

This classical description successfully explains the low-field Hall response observed in Sec. 4.2 and motivates the use of the low-field Hall slope as one independent estimate of the carrier density. However, it fails to account for the quantised plateaux and vanishing longitudinal resistance observed at higher magnetic fields.

2.2 Landau Quantisation

In a strong perpendicular magnetic field, the classical picture of continuous electronic energy bands breaks down. Instead, the cyclotron motion of charge carriers becomes quantised, leading to discrete energy levels known as Landau levels [3].

For a two-dimensional electron gas, the Landau level energies are given by

$$E_n = \hbar\omega_c \left(n + \frac{1}{2}\right), \tag{2}$$

where n is a non-negative integer and $\omega_c = eB/m^*$ is the cyclotron frequency. The spacing between Landau levels therefore increases linearly with magnetic field.

As the magnetic field is increased, the number of occupied Landau levels decreases, and the electronic system can be characterised by an integer filling factor ν, defined as the number of occupied Landau levels [3]. This quantisation of the energy spectrum underlies the appearance of discrete features in magnetotransport measurements.

The concept of a filling factor provides the basis for assigning integer values of ν to the Hall plateaux observed in Sec. 4.3, where the $\nu = 2$ plateau is identified from the measured Hall resistance.

2.3 Density of States and Disorder Broadening

In the absence of disorder, Landau quantisation leads to a density of states consisting of a series of delta-function-like peaks at the Landau level energies [3]. In real materials, however, impurities, interface roughness, and other sources of disorder broaden these levels.

Disorder transforms each Landau level into a broadened band containing both localised and extended electronic states [3]. States near the centres of the broadened Landau levels are typically extended and contribute to electrical conduction, while states in the tails are localised and do not carry current over macroscopic distances.

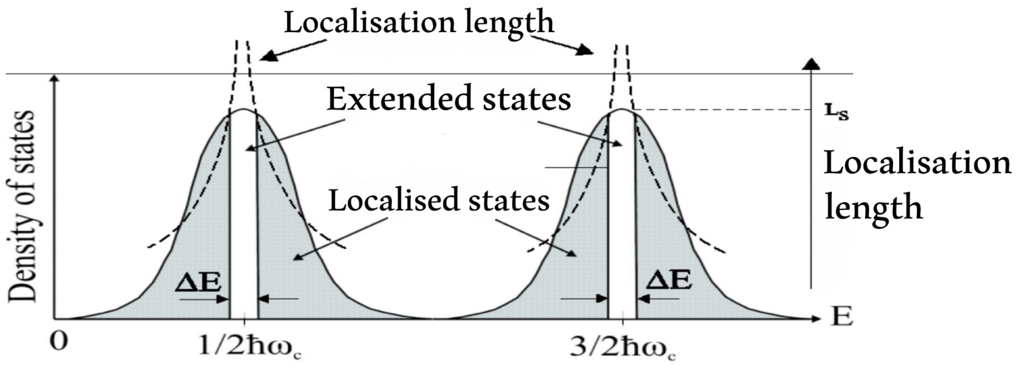

Figure 2: Schematic illustration of Landau quantisation, disorder-induced broadening, and localisation. Discrete Landau levels broaden into bands containing extended states near their centres and localised states in the tails. As the Fermi level moves through localised regions, Hall plateaux form while longitudinal transport is suppressed. Reproduced from [3].

As the magnetic field is varied, the Fermi level moves through these broadened Landau levels. When the Fermi level lies in a region dominated by localised states, the Hall resistance remains constant while longitudinal transport is suppressed [7].

This localisation-based picture explains the formation of Hall plateaux and predicts the correlated behaviour observed experimentally, namely plateaux in R_{xy} accompanied by strong suppression of R_{xx}, as demonstrated in Sec. 4.3 and illustrated schematically in Fig. 2.

2.4 Shubnikov–de Haas Oscillations

Before fully developed Hall plateaux appear, Landau quantisation manifests experimentally as oscillations in the longitudinal resistance, known as Shubnikov–de Haas oscillations [5]. These oscillations arise from the periodic modulation of the density of states at the Fermi level as successive Landau levels cross it.

Shubnikov–de Haas oscillations are periodic in inverse magnetic field, with a period determined by the carrier density [5]. As a result, analysis of their periodicity provides a quantum-mechanical method for extracting the carrier density independently of the classical Hall slope.

In this work, both the periodicity of the Shubnikov–de Haas oscillations and a Landau fan analysis are used to extract n_s, as presented in Sec. 4.2. The appearance of these oscillations therefore serves as a precursor signature of the quantum Hall regime.

2.5 Edge-State Transport

An alternative and complementary interpretation of quantum Hall transport is provided by the edge-state picture [8]. In a strong magnetic field, the confining potential at the boundaries of a two-dimensional electron gas gives rise to chiral edge channels that propagate along the sample edges, as illustrated schematically in Fig. 3.

Because these edge channels are spatially separated and propagate in a single direction, backscattering is strongly suppressed [9]. As a result, longitudinal dissipation is minimised, and the longitudinal resistance approaches zero when transport is dominated by edge channels.

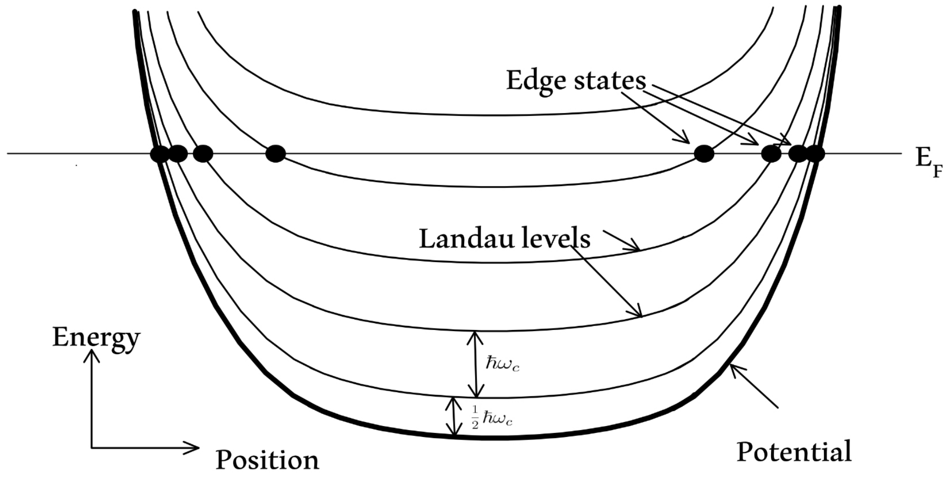

Figure 3: Schematic illustration of edge-state transport in the integer quantum Hall regime. Chiral edge channels propagate along the sample boundaries with suppressed backscattering, leading to dissipationless longitudinal transport. Adapted from [8].

While the present experiment does not directly probe edge transport, the strong suppression of R_{xx} observed on the $\nu = 2$ plateau is consistent with the edge/localisation framework commonly used to interpret the integer quantum Hall effect.

2.6 Resistivity and Conductivity Tensors

Experimental measurements of magnetotransport are naturally expressed in terms of resistances, whereas theoretical descriptions often employ resistivity and conductivity tensors [7]. In two dimensions, the resistivity tensor is characterised by longitudinal and Hall components, ρ_{xx} and ρ_{xy}.

The conductivity tensor, defined as the inverse of the resistivity tensor, provides a particularly clear description of quantum Hall transport. In the integer quantum Hall regime, plateaux in σ_{xy} occur at integer multiples of e^2/h, while σ_{xx} exhibits minima corresponding to suppressed longitudinal conduction [7].

This tensor formulation provides a direct link between the localisation picture and experimental observables. The conversion from measured resistances to resistivity and conductivity tensor components, and the comparison of their behaviour, is carried out in Sec. 4.4.

2.7 Relevant Energy Scales

Observation of quantum Hall phenomena requires that the discrete structure of the Landau levels be resolved [3, 7]. This condition is governed by the competition between several energy scales, principally the Landau level spacing, thermal broadening, and disorder-induced broadening.

The spacing between Landau levels is given by

$$\Delta E = \hbar\omega_c = \frac{\hbar e B}{m^*}, \tag{3}$$

which increases linearly with magnetic field [10]. To resolve individual Landau levels, this spacing must exceed the energy scales associated with both temperature and disorder.

2.8 Thermal Broadening

Finite temperature introduces thermal broadening of the electronic occupation around the Fermi level, characterised by the thermal energy scale $k_B T$ [5]. When $k_B T$ becomes comparable to the Landau level spacing, the sharp modulation of the density of states is smeared, suppressing both Shubnikov–de Haas oscillations and Hall quantisation [5].

Resolving quantum oscillations and Hall plateaux therefore requires operation in the regime

$$k_B T \ll \hbar\omega_c.$$

This inequality provides the fundamental motivation for performing quantum Hall measurements at cryogenic temperatures [7].

2.9 Disorder Broadening

In addition to thermal effects, disorder broadens Landau levels by introducing spatial variations in the electronic potential [3]. This broadening gives rise to regions of localised and extended states within each Landau level, as discussed in Sec. 2.

The finite width of Hall plateaux reflects the combined effects of disorder-induced broadening and finite temperature [7]. At higher temperatures or in more disordered samples, extended states overlap more strongly, reducing plateau width and limiting the number of observable filling factors.

2.10 Experimental Implications

The requirement that thermal and disorder broadening remain smaller than the Landau level spacing dictates the experimental conditions necessary to observe the integer quantum Hall effect [7]. Specifically, low temperatures are required to suppress thermal smearing, while sufficiently high magnetic fields are needed to increase the Landau level spacing.

These energy-scale considerations directly motivate the use of a cryogenic environment, liquid helium cooling, and a superconducting magnet in the present experiment. They also determine the accessible range of filling factors and provide a natural explanation for the temperature dependence of the Shubnikov–de Haas oscillations analysed in Sec. 4.6.

3 Experimental Methods

3.1 Sample, Geometry, and Contacts

Magnetotransport measurements were performed on a GaAs/AlGaAs two-dimensional electron gas patterned into a Hall-bar geometry, enabling simultaneous determination of the longitudinal and Hall resistances, R_{xx} and R_{xy}, under identical experimental conditions. The Hall bar provides well-defined current and voltage probe locations so that four-terminal voltages can be measured with minimal sensitivity to lead resistances.

The device geometry was characterised by the ratio W/L, where W is the channel width between the longitudinal voltage probes and L is the separation of those probes along the current path. This geometric factor is required to convert measured resistances into resistivities via $\rho_{xx} = (W/L)\, R_{xx}$, and its uncertainty contributes a systematic component to the uncertainty budget in the extracted mobility. In this work, W/L was taken to be $W/L = 0.061 \pm 0.009$, estimated from an optical image of the Hall bar, as described in Appendix D.

Electrical contact to the Hall bar was made using bonded wires connected to labelled terminals corresponding to current injection, ground return, and voltage probe pairs. During initial setup, intermittent or high-resistance continuity on several breakout-box to cryostat contact lines was observed and mitigated by selecting an alternate set of functional contact lines that preserved the Hall-bar measurement geometry, after which the contact resistances remained stable over the measurement campaign.

3.2 Cryogenics and Magnet System

To access the regime required for observation of quantum Hall transport, the sample was cooled in a helium cryostat to temperatures in the range $T = 3\,\mathrm{K}$ to $5\,\mathrm{K}$ and subjected to a perpendicular magnetic field generated by a superconducting solenoid. Cooling suppresses thermal broadening of Landau levels, while high magnetic field increases the Landau level spacing, together enabling resolution of quantised Hall plateaux and Shubnikov–de Haas oscillations.

The cryogenic system comprised a laboratory helium insert cryostat inserted into a liquid-helium dewar, with sample temperature monitored using a cryogenic temperature sensor mounted near the sample stage. A leak in the cryostat vacuum space prevented stable operation at the nominal base temperature of the system; to maintain thermal stability, an active thermal stabilisation configuration was used, limiting the lowest reproducible temperature to approximately $3\,\mathrm{K}$. This temperature limitation reduces the visibility of higher filling-factor features but does not prevent observation of the defining $\nu = 2$ quantum Hall signatures reported in Sec. 4.

Magnetic field sweeps were performed over the range $B = 0\,\mathrm{T}$ to $5\,\mathrm{T}$ using the superconducting magnet power supply (laboratory superconducting magnet power supply), with both up-sweep and down-sweep traces recorded to assess sweep-to-sweep reproducibility. The magnetic field value reported in this work corresponds to the power-supply readback using the laboratory conversion factor $B = (0.13445\,\mathrm{T/A})\, I$, and the same calibration was applied consistently across all analysis methods relying on B (Hall slope, SdH periodicity, and filling-factor assignment).

3.3 Electronics and Measurement Configuration

Longitudinal and Hall voltages were measured using lock-in detection to maximise signal-to-noise at the small excitation currents required to avoid electron heating at cryogenic temperatures. An AC excitation current of amplitude $I \approx 1\,\mu\mathrm{A}$ at frequency $f = 67\,\mathrm{Hz}$ was driven through the Hall bar using the SR830 lock-in internal sine output in series with a $1\,\mathrm{M\Omega}$ resistor, while the resulting V_{xx} and V_{xy} signals were measured with two Stanford Research Systems SR830 lock-in amplifiers referenced to the excitation frequency.

Signal integrity was improved by lock-in detection with differential (A–B) inputs and a shared reference between instruments along the cryostat wiring loom, reducing pickup from environmental noise and suppressing high-frequency components that can contribute to measurement instability. To minimise systematic offsets, voltage channels were configured to measure differential potentials between the appropriate probe pairs, and the

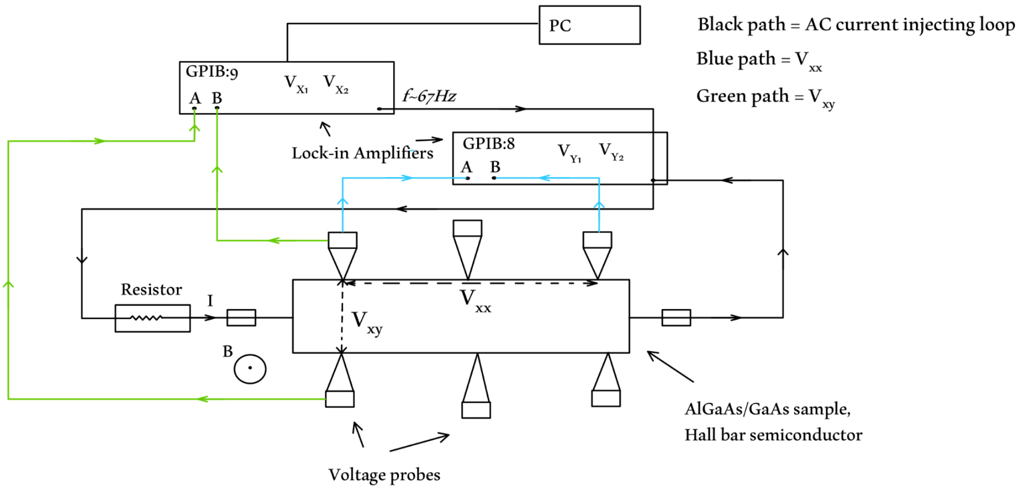

Figure 4: Hall-bar geometry and measurement wiring configuration. The diagram shows the current injection path, longitudinal and transverse voltage probe pairs, and the lock-in detection scheme used to extract $R_{xx} = V_{xx}/I$ and $R_{xy} = V_{xy}/I$.

measurement polarity conventions for V_{xx} and V_{xy} were kept consistent across all temperature and field sweeps.

Fig. 4 summarises the measurement configuration, showing the Hall-bar contact arrangement, current injection path, and voltage probe pairs used for the extraction of $R_{xx} = V_{xx}/I$ and $R_{xy} = V_{xy}/I$. This configuration allows R_{xx} and R_{xy} to be determined simultaneously as functions of magnetic field and temperature, ensuring that the longitudinal and transverse responses are directly comparable.

3.4 Data Acquisition, Corrections, and Analysis Pipeline

For each field sweep, time-stamped measurements of magnetic field B, temperature T, and lock-in voltages V_{xx} and V_{xy} were recorded at a sampling interval of 1 s and subsequently converted to resistances using the measured or calibrated operating current. All data processing, figure generation, and parameter extraction were performed using Python scripts developed for this project.

A constant scaling offset of approximately 5 % was identified in one Hall-voltage channel due to an instrument configuration setting and corrected prior to parameter extraction. The correction was applied uniformly across all affected datasets and does not alter qualitative transport trends; instead, it improves agreement between independent carrier-density estimates obtained from classical Hall slope and quantum oscillation analyses. All reported transport parameters were extracted from the corrected data.

Uncertainties were estimated by combining statistical contributions from fits or peak-finding procedures with systematic contributions from instrument scaling and the Hall-bar geometry factor. In particular, the uncertainty in W/L dominates the uncertainty in quantities derived from ρ_{xx} (such as mobility), whereas carrier-density estimates derived from R_{xy} slopes and oscillation periodicities are primarily limited by the precision of the magnetic-field assignment and the plateau/feature identification criteria. Additional validation of peak detection and smoothing choices is provided in Appendix A.

3.5 Operating Current Calibration from the $\nu = 2$ Plateau

In principle, the quantised Hall resistance associated with a well-resolved $\nu = 2$ plateau can provide an absolute reference that may be used to determine the effective excitation current flowing through the sample. This approach is independent of instrument readback and exploits the robustness of the quantised resistance value.

Accordingly, the operating current can be estimated using the median Hall voltage measured within the $\nu = 2$ plateau window via the relation

$$I = \frac{V_{xy}}{R_{xy}}, \tag{4}$$

where R_{xy} is taken to be the ideal quantised value $h/2e^2$. Applying this procedure yields a plateau-derived current of

$$I = (8.35 \pm 0.15) \times 10^{-7}\,\mathrm{A},$$

with an uncertainty dominated by the spread of Hall voltages within the plateau window and instrumental voltage noise.

For comparison, the excitation current inferred from the lock-in amplifier readback was approximately

$$I_{\text{lock-in}} \approx (7.1 \pm 0.2) \times 10^{-7}\,\mathrm{A},$$

corresponding to a discrepancy of approximately 17 % between the two estimates.

At this stage of the analysis, both the nominal lock-in readback current and the plateau-derived current are retained for comparison, with no preference assigned. The adoption of the plateau-derived current for downstream quantitative analysis is deferred until after the experimental identification and characterisation of the $\nu = 2$ Hall plateau.

4 Results

4.1 Overview of Magnetotransport

We begin by presenting an overview of the longitudinal and Hall transport measured at the base temperature of the experiment, $T = 3\,\mathrm{K}$. This overview establishes that the sample is in the quantum transport regime and exhibits the characteristic signatures required for observation of the integer quantum Hall effect using the nominal operating current defined in Sec. 3.5.

Fig. 5 shows the Hall resistance R_{xy} and longitudinal resistance R_{xx} measured simultaneously as functions of perpendicular magnetic field. The Hall resistance exhibits a step-like increase with increasing field, while the longitudinal resistance displays oscillatory behaviour that becomes increasingly pronounced at higher magnetic fields.

The oscillations observed in R_{xx} are identified as Shubnikov–de Haas oscillations, arising from the magnetic-field-induced modulation of the density of states at the Fermi level. Their presence indicates that Landau quantisation is resolved in this sample under the present experimental conditions.

At higher magnetic fields, extended regions of approximately constant R_{xy} are accompanied by strong suppression of R_{xx}. This correlated behaviour is the defining experimental signature of the integer quantum Hall effect and demonstrates that the sample and measurement configuration are suitable for quantitative analysis of quantum Hall transport.

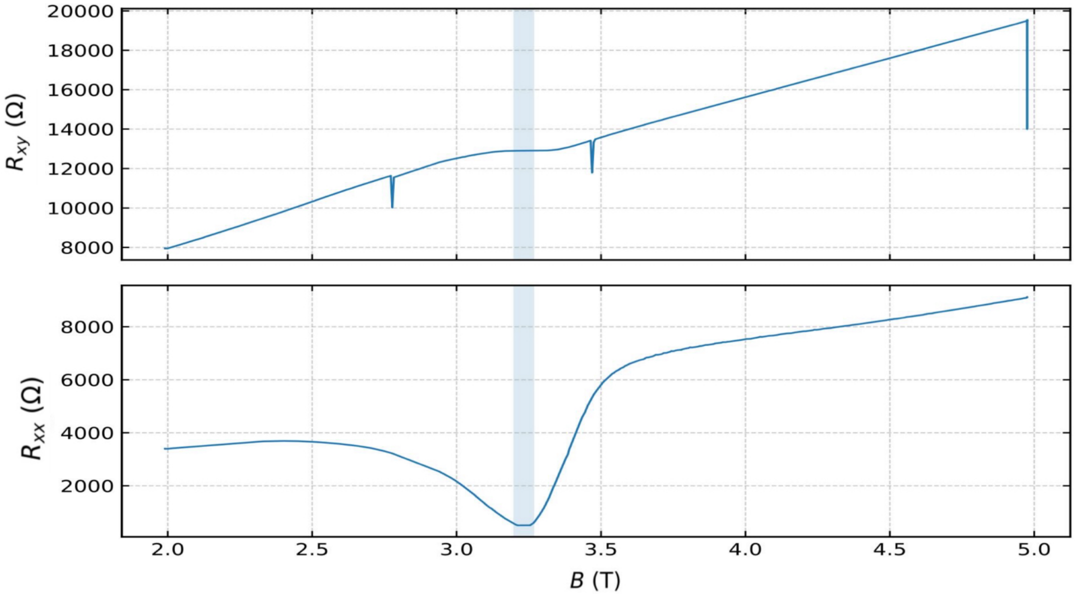

Figure 5: Overview of magnetotransport at $T = 3\,\mathrm{K}$. The Hall resistance R_{xy} exhibits a staircase-like dependence on magnetic field, while the longitudinal resistance R_{xx} displays Shubnikov–de Haas oscillations that increase in amplitude with field. The coexistence of Hall plateaux and oscillatory longitudinal transport indicates that the system is in the quantum transport regime.

4.2 Carrier Density Determination

The two-dimensional carrier density n_s was determined using four independent methods probing both classical and quantum transport regimes. Agreement between these methods provides a stringent internal consistency check and supports the interpretation of transport as arising from a single, well-defined carrier population.

The carrier density was first estimated from the low-field Hall slope, using the classical relation

$$R_{xy} = \frac{B}{n_s e}, \tag{5}$$

where the slope of $R_{xy}(B)$ at small magnetic fields yields n_s. This method is insensitive to quantum oscillations and provides a baseline classical estimate of the carrier density.

Independent quantum estimates of n_s were obtained from the periodicity of Shubnikov–de Haas oscillations in the longitudinal resistance. Because these oscillations are periodic in inverse magnetic field, their spacing $\Delta(1/B)$ is related to the carrier density via

$$n_s = \frac{2e}{h\,\Delta(1/B)} \tag{6}$$

where the factor of two accounts for spin degeneracy. This method probes the same carrier population through Landau quantisation of the density of states through the filling-factor condition.

A complementary quantum determination of n_s was obtained from a Landau fan analysis, in which the indices of Shubnikov–de Haas extrema were plotted as a function of inverse magnetic field. A Landau fan diagram plots the index of Shubnikov–de Haas extrema as a function of inverse magnetic field. The slope of the resulting linear relation yields the carrier density independently of the absolute magnetic-field offset. The corresponding Landau fan diagram is shown in Appendix C3.

Finally, the carrier density was extracted from the magnetic-field position of the $\nu = 2$ Hall plateau using the classical relation in Eq. (5), where B_ν is the magnetic field at which the filling factor ν is realised. This method directly links the carrier density to the quantised Hall response observed at higher magnetic fields.

Table 1 summarises the carrier densities extracted using all four methods. The values cluster tightly around a mean carrier density of

$$n_s = (1.57 \pm 0.01) \times 10^{15}\,\mathrm{m}^{-2},$$

with a maximum fractional spread of approximately 0.3 %. The close agreement between classical and quantum determinations indicates that a single carrier population governs transport across the full magnetic-field range and validates the consistency of the magnetic-field calibration and filling-factor assignment.

Table 1: Carrier density values extracted using four independent methods probing both classical and quantum transport regimes. Quoted uncertainties reflect the spread associated with each extraction method as determined from plateau statistics, SdH peak spacing, Landau fan fitting, and low-field Hall slope analysis [11].

Method	n_s ($\times 10^{15}\,\mathrm{m}^{-2}$)
$\nu = 2$ Hall plateau	1.563 ± 0.005
SdH periodicity $\Delta(1/B)$	1.574 ± 0.004
Landau fan analysis	1.572 ± 0.003
Low-field Hall slope	1.562 ± 0.006
Mean value adopted	1.57 ± 0.01

4.3 Hall Resistance Quantisation and the $\nu = 2$ Plateau

For the purposes of establishing the existence and robustness of the $\nu = 2$ Hall plateau, Hall resistance values presented in this subsection are calculated using the nominal excitation current inferred from the lock-in amplifier readback. At this stage, no use is made of plateau-based current calibration, ensuring that the identification and characterisation of the plateau are independent of any subsequent refinement of the operating current.

We now examine the quantisation of the Hall resistance in detail, focusing on the most clearly resolved plateau corresponding to filling factor $\nu = 2$. This plateau is expected to be the most robust under conditions of finite temperature and moderate disorder and therefore provides a stringent test of quantised Hall transport in the present apparatus.

Fig. 6 shows a magnified view of the Hall resistance $R_{xy}(B)$ in the magnetic-field region where the $\nu = 2$ plateau is observed. The plateau is centred at a magnetic field of

$$B_{\nu=2} = (3.23 \pm 0.01)\,\mathrm{T},$$

and extends over the field interval

$$3.19\,\mathrm{T} \leq B \leq 3.27\,\mathrm{T},$$

within which the Hall resistance remains approximately constant.

Within this plateau window, the median measured Hall resistance is

$$R_{xy} = (12905.9 \pm 0.3)\,\Omega,$$

which differs from the ideal quantised value

$$\frac{h}{2e^2} = 12\,906.4\,\Omega$$

by tens of parts per million. This level of agreement demonstrates clear Hall resistance quantisation on the $\nu = 2$ plateau.

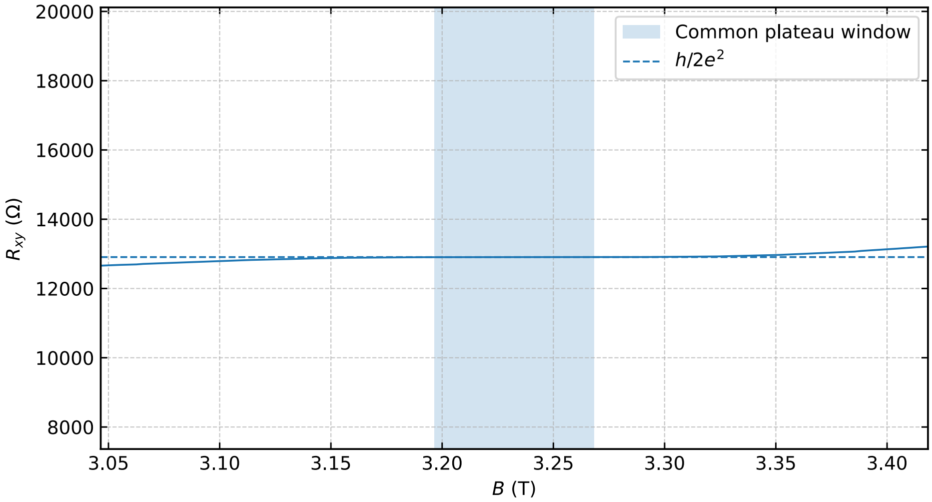

Figure 6: Magnified view of the Hall resistance $R_{xy}(B)$ in the vicinity of the $\nu = 2$ plateau. The shaded region indicates the plateau window over which the Hall resistance remains approximately constant and close to the ideal quantised value $h/2e^2$.

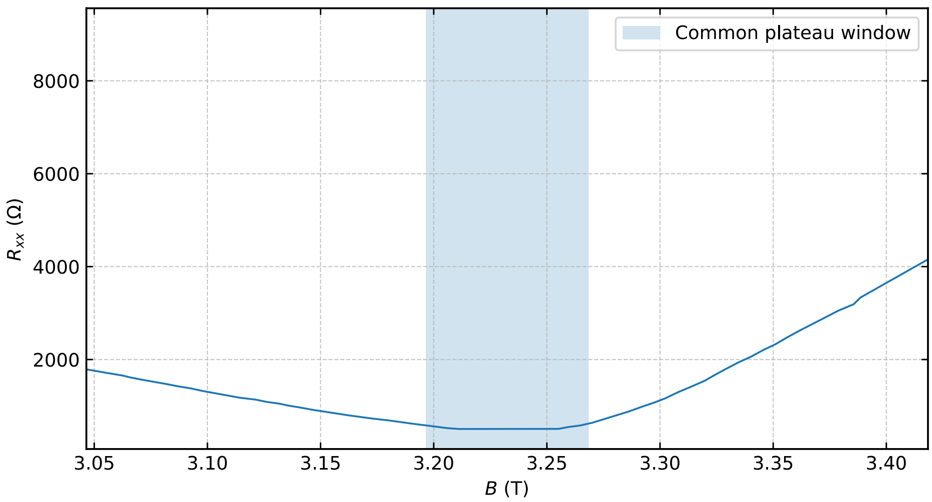

Figure 7: Longitudinal resistance $R_{xx}(B)$ in the vicinity of the $\nu = 2$ plateau. Strong suppression of R_{xx} is observed over the same magnetic-field interval as the Hall plateau, indicating dissipationless transport.

The corresponding longitudinal resistance in the same magnetic-field range is shown in Fig. 7. Over the plateau window identified above, R_{xx} is strongly suppressed and approaches zero within the measurement resolution. The simultaneous observation of a quantised R_{xy} plateau and suppressed R_{xx} constitutes the defining experimental signature of the integer quantum Hall effect.

To assess the robustness of the observed plateau, Hall resistance traces acquired during both increasing and decreasing magnetic-field sweeps were compared. Fig. 8 shows the deviation of R_{xy} from the ideal quantised value $h/2e^2$ for up-sweep and down-sweep data. The two traces agree within the quoted deviation, with no evidence of significant hysteresis or sweep-direction-dependent effects.

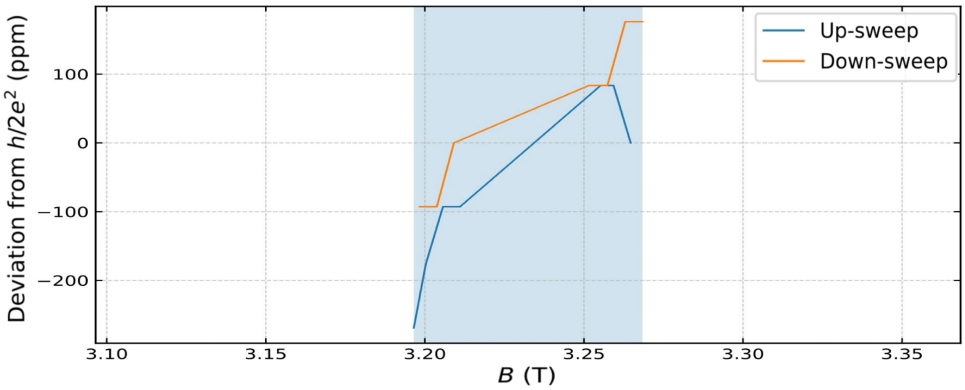

Figure 8: Deviation of the Hall resistance from the ideal quantised value $h/2e^2$ across the $\nu = 2$ plateau for increasing and decreasing magnetic-field sweeps. The close agreement between traces demonstrates sweep-to-sweep reproducibility and negligible hysteresis.

The consistency of the $\nu = 2$ plateau position, width, and quantised resistance value across sweep directions indicates that the observed quantisation is not an artefact of transient heating or sweep-dependent instabilities. This robustness supports the use of the $\nu = 2$ plateau as a reliable reference feature for subsequent parameter extraction.

Having established the existence, reproducibility, and robustness of the $\nu = 2$ Hall plateau independently of any plateau-based calibration, the quantised Hall resistance may now be used as a physical reference to refine the effective excitation current. This refinement, described in Sec. 3.5, is therefore applied only after the plateau has been experimentally identified and validated, avoiding any circular dependence between current calibration and plateau quantisation.

4.4 Resistivity and Conductivity Tensor Analysis

To facilitate comparison with theoretical descriptions of quantum Hall transport, the measured longitudinal and Hall resistances were converted into resistivity and conductivity tensor components. This analysis allows the relationship between dissipationless transport, localisation, and Hall quantisation to be examined in a form commonly used in the quantum Hall literature.

The longitudinal and Hall resistivities were obtained from the measured resistances using the Hall-bar geometry factor,

$$\rho_{xx} = \frac{W}{L} R_{xx}, \qquad \rho_{xy} = R_{xy}, \tag{7}$$

where $W/L = 0.061 \pm 0.009$. The uncertainty in this geometric factor constitutes the dominant systematic contribution to the uncertainty in ρ_{xx} and in quantities derived from it, such as the mobility.

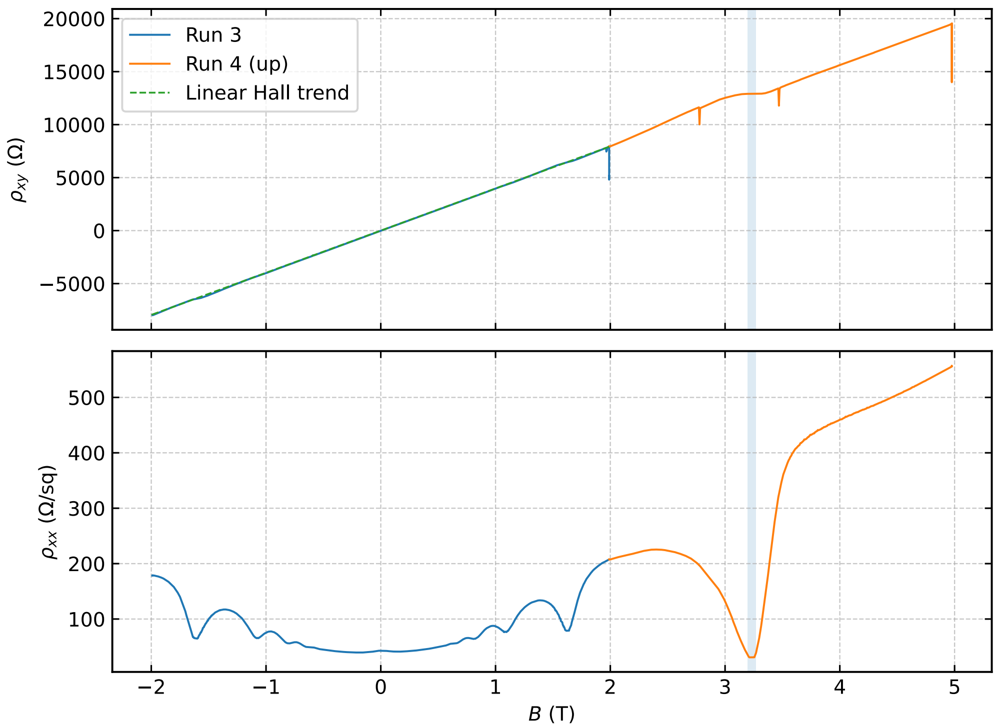

Figure 9: Resistivity tensor components $\rho_{xx}(B)$ and $\rho_{xy}(B)$. At low magnetic field, ρ_{xy} follows the classical Hall relation, while at higher fields it develops plateaux accompanied by strong suppression of ρ_{xx}, indicating the transition to quantised Hall transport.

Fig. 9 shows the resulting resistivity tensor components $\rho_{xx}(B)$ and $\rho_{xy}(B)$. At low magnetic fields, ρ_{xy} increases approximately linearly with B, consistent with classical Hall behaviour, while ρ_{xx} remains finite. At higher magnetic fields, ρ_{xy} develops plateaux coincident with strong suppression of ρ_{xx}, reflecting the onset of quantised Hall transport.

The conductivity tensor components σ_{xx} and σ_{xy} were obtained by inversion of the resistivity tensor according to

$$\begin{aligned}\sigma_{xx} &= \frac{\rho_{xx}}{\rho_{xx}^2 + \rho_{xy}^2},\\ \sigma_{xy} &= \frac{\rho_{xy}}{\rho_{xx}^2 + \rho_{xy}^2}.\end{aligned} \tag{8}$$

[3, 7] This standard inversion follows the conventional tensor treatment of magneto-transport in two-dimensional systems [7]. Fig. 10 displays the resulting conductivity components as functions of magnetic field.

As the magnetic field approaches zero, the Hall resistivity ρ_{xy} also approaches zero, causing the denominator $\rho_{xx}^2 + \rho_{xy}^2$ to become small. In this regime, the tensor inversion becomes numerically sensitive to noise and small uncertainties in ρ_{xx} and ρ_{xy}. Conductivity values in the immediate vicinity of $B = 0$ are therefore ill-conditioned and are treated with caution.

To avoid over-interpretation of numerically unstable data, the low-field region near $B = 0$ was excluded from quantitative analysis of the conductivity tensor and is shown as a masked region in Fig. 10. Outside this region, clear minima in σ_{xx} coincide with plateaux in σ_{xy}, consistent with localisation of bulk states between Landau levels and the transport picture underlying the integer quantum Hall effect.

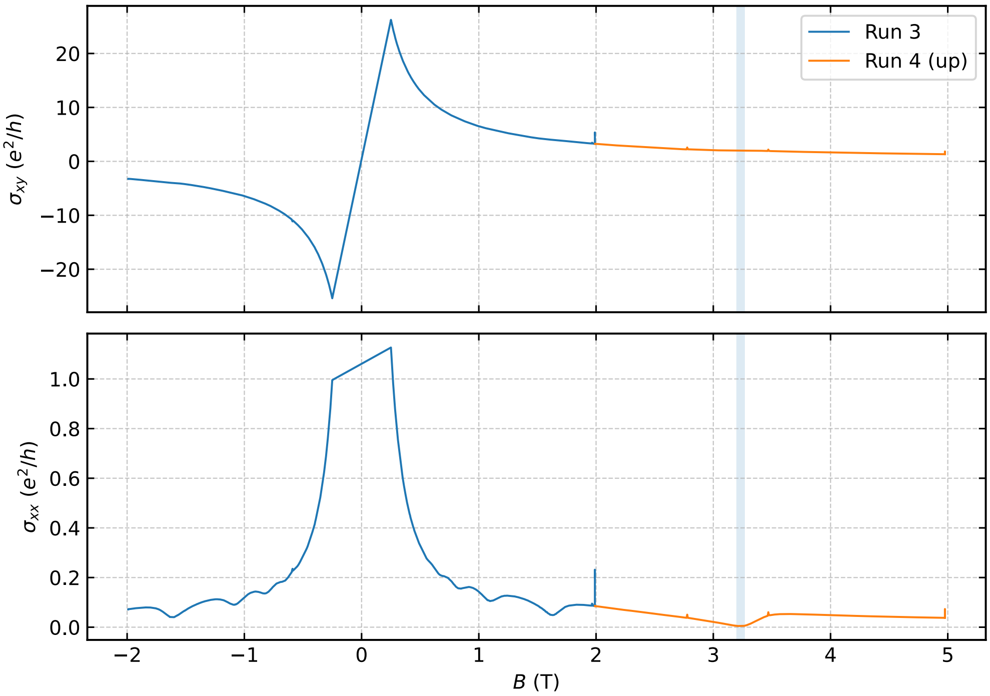

Figure 10: Conductivity tensor components $\sigma_{xx}(B)$ and $\sigma_{xy}(B)$ obtained by inversion of the resistivity tensor. The low-field region near $B = 0$ is masked due to numerical instability of the inversion. Outside this region, minima in σ_{xx} coincide with plateaux in σ_{xy}.

4.5 Longitudinal Transport and Mobility

The low-field longitudinal transport was analysed to extract the carrier mobility, which provides a quantitative measure of sample quality and the degree of disorder present in the two-dimensional electron gas. The mobility also sets the characteristic field scale at which Shubnikov–de Haas oscillations and quantum Hall features are expected to emerge.

At sufficiently low magnetic fields, where quantum oscillations are absent, the longitudinal resistivity is related to the carrier density and mobility through

$$\rho_{xx}(B \to 0) = \frac{1}{n_s e \mu} \tag{9}$$

Using the carrier density determined in Sec. 4.2 and Eq. (9), the mobility was extracted from the measured low-field ρ_{xx} values.

Fig. 11 shows the low-field diagnostics used to determine $\rho_{xx}(B \to 0)$ and assess the stability of the extracted mobility. The resulting mobility is

$$\mu = (9.5 \pm 1.4) \times 10^5\,\mathrm{cm^2\,V^{-1}\,s^{-1}}.$$

Using an effective electron mass of $m^* = 0.067\,m_e$ for GaAs, the transport lifetime was estimated from the relation

$$\tau_{\mathrm{tr}} = \frac{m^* \mu}{e} \tag{10}$$

yielding

$$\tau_{\mathrm{tr}} \approx 3.5 \times 10^{-11}\,\mathrm{s},$$

with uncertainty dominated by that of the mobility.

The dominant contribution to the uncertainty in the extracted mobility arises from the Hall-bar geometry factor W/L, rather than from voltage noise or fitting uncertainty. This indicates that systematic geometric effects, rather than electronic instability, limit the precision of the mobility determination in the present experiment.

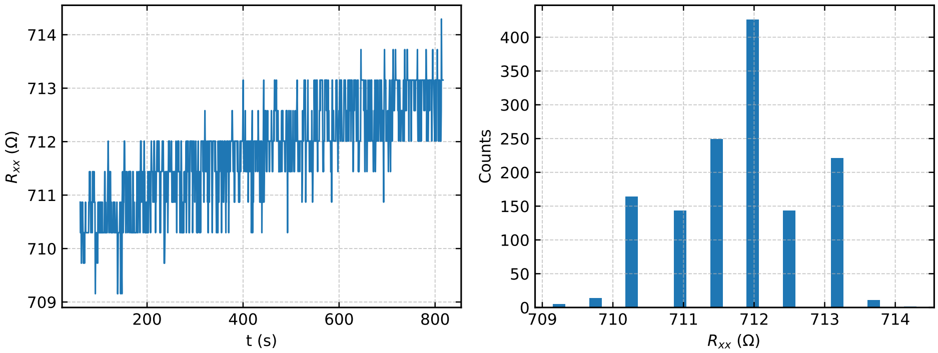

Figure 11: Low-field longitudinal transport diagnostics used to extract the carrier mobility. The stability (left) and consistency (right) of the low-field ρ_{xx} values enable reliable determination of $\rho_{xx}(B \to 0)$ and hence the mobility.

4.6 Temperature Dependence of Shubnikov–de Haas Oscillations

Fig. 12 shows the longitudinal voltage $V_{xx}(B)$ measured at the three temperatures over the same magnetic-field range. As the temperature increases, the amplitude of the Shubnikov–de Haas oscillations decreases monotonically, while the overall field positions of the oscillations remain unchanged.

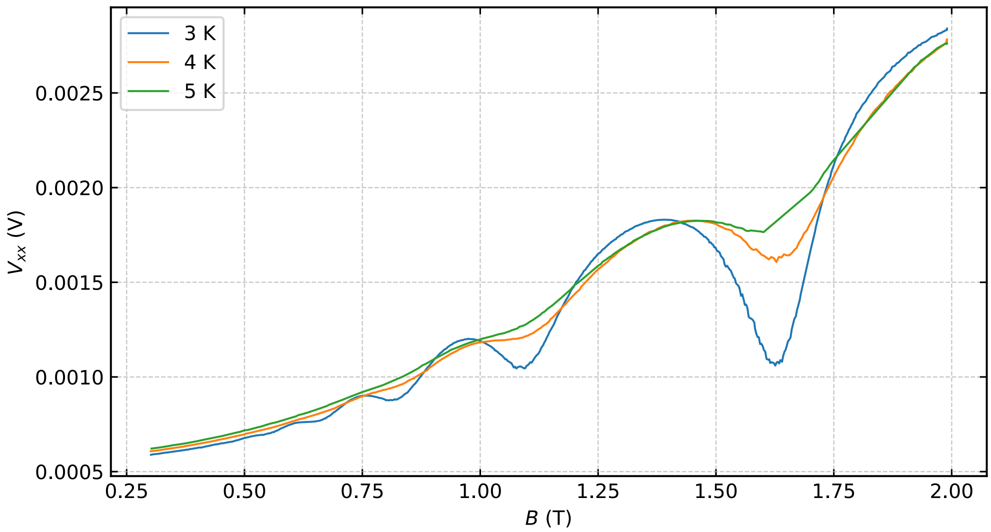

Figure 12: Longitudinal voltage $V_{xx}(B)$ measured at $T = 3\,\mathrm{K}$, $4\,\mathrm{K}$, and $5\,\mathrm{K}$. The amplitude of the Shubnikov–de Haas oscillations decreases monotonically with increasing temperature, while the oscillation positions remain unchanged.

The reduction in oscillation amplitude with increasing temperature is consistent with thermal broadening of Landau levels, which suppresses the oscillatory modulation of the density of states at the Fermi level. This behaviour is expected when the thermal energy scale $k_B T$ becomes comparable to the Landau level spacing.

The temperature dependence of the longitudinal resistance was examined to assess the influence of thermal broadening on quantum oscillations and to test the energy-scale arguments outlined in Sec. 3. Measurements were performed at temperatures of $T = 3\,\mathrm{K}$, $4\,\mathrm{K}$, and $5\,\mathrm{K}$, spanning the stable operating range of the cryogenic system.

To quantify this trend, the oscillation amplitude was extracted at a fixed magnetic field for each temperature and is plotted as a function of temperature in Fig. 13. Because data are available at only three temperatures, the analysis is restricted to qualitative trends rather than a full Lifshitz–Kosevich fit.

The low-field Hall slope was found to be unchanged across the measured temperature range, indicating that the carrier density remains constant between $3\,\mathrm{K}$ and $5\,\mathrm{K}$. This confirms that the observed temperature dependence of the oscillations arises from thermal broadening rather than from changes in carrier density.

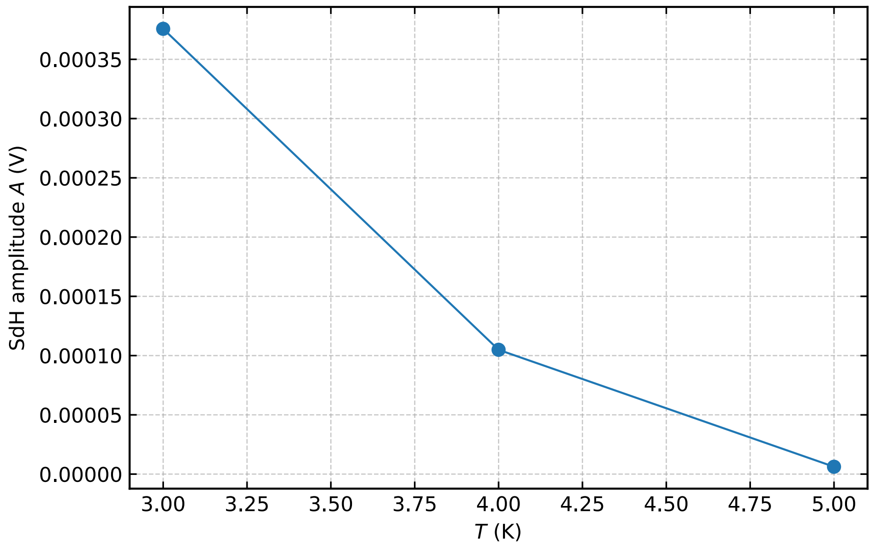

Figure 13: Extracted Shubnikov–de Haas oscillation amplitude as a function of temperature. The monotonic decrease in amplitude reflects thermal broadening of Landau levels. The limited number of temperature points restricts the analysis to qualitative trends.

4.7 Summary of Final Transport Parameters

Table 2 summarises the final transport parameters extracted in this work, together with the key experimental conditions under which they were obtained. These values are compiled from the analyses presented in Secs. 4.1–4.6 and represent the quantitative basis for the discussion in Sec. 5.

Table 2: Summary of final transport parameters extracted from magnetotransport measurements on the GaAs/AlGaAs two-dimensional electron gas. Uncertainties represent dominant experimental contributions as discussed in Sec. 5.

Quantity	**Value**
Carrier density, n_s	$(1.57 \pm 0.01) \times 10^{15}\,\mathrm{m}^{-2}$
Carrier density agreement	$\sim 0.3\,\%$ (four independent methods)
Filling factor analysed	$\nu = 2$
Magnetic field at $\nu = 2$	$(3.23 \pm 0.01)\,\mathrm{T}$
Hall resistance on plateau	$R_{xy} = (12905.9 \pm 0.3)\,\Omega$
Deviation from $h/2e^2$	Tens of parts per million
Operating current (plateau-derived)	$(8.35 \pm 0.15) \times 10^{-7}\,\mathrm{A}$
Lock-in readback current	$(7.1 \pm 0.2) \times 10^{-7}\,\mathrm{A}$
Mobility, μ	$(9.5 \pm 1.4) \times 10^{5}\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
Transport lifetime, τ_{tr}	$\approx 3.5 \times 10^{-11}\,\mathrm{s}$
Temperature range	$3\,\mathrm{K}$ to $5\,\mathrm{K}$

5 Discussion

5.1 Demonstration of the Integer Quantum Hall Effect

The results presented in Sec. 5 demonstrate the defining experimental signatures of the integer quantum Hall effect in the GaAs/AlGaAs two-dimensional electron gas studied in this work. These signatures include the appearance of a quantised Hall resistance plateau at an integer filling factor, accompanied by strong suppression of the longitudinal resistance.

Specifically, a well-defined plateau is observed at filling factor $\nu = 2$, centred at $B_{\nu=2} = 3.23\,\mathrm{T}$, with the measured Hall resistance deviating from the ideal quantised value $h/2e^2$ by only tens of parts per million. Over the same magnetic-field interval, the longitudinal resistance R_{xx} is strongly suppressed, satisfying the canonical transport criteria for the integer quantum Hall effect.

The agreement between up-sweep and down-sweep measurements further indicates that the observed quantisation is robust against sweep-direction-dependent effects and transient heating. Taken together, these observations demonstrate that the essential physics of the integer quantum Hall effect is realised in the present experimental apparatus, despite its operation under laboratory-scale constraints.

5.2 Connection to the Localisation and Edge-State Picture

The observed transport behaviour is consistent with the localisation-based interpretation of the integer quantum Hall effect described in Sec. 2. In this picture, disorder broadens the Landau levels, leading to localised bulk states between extended states at the centres of the levels.

As the magnetic field is varied, the Fermi level traverses regions of localised states, producing plateaux in the Hall resistance while suppressing longitudinal dissipation. This interpretation is supported experimentally by the coincidence of minima in σ_{xx} with plateaux in σ_{xy} observed in the conductivity tensor analysis.

The strong suppression of R_{xx} on the $\nu = 2$ plateau is also consistent with the edge-state transport framework, in which current is carried by chiral edge channels with suppressed backscattering. While the present measurements do not directly probe edge transport, the observed dissipationless behaviour aligns with the expectations of the edge/localisation picture underlying the integer quantum Hall effect [3, 7].

5.3 Carrier Density, Mobility, and Sample Quality

The carrier density extracted in this work, $n_s = 1.57 \times 10^{15}\,\mathrm{m}^{-2}$, lies within the typical range reported for modulation-doped GaAs/AlGaAs two-dimensional electron gases, which commonly exhibit densities of order 10^{15}–$10^{16}\,\mathrm{m}^{-2}$ [3]. The close agreement between carrier densities obtained from four independent methods indicates that transport is dominated by a single, well-defined carrier population, with no evidence for parallel conduction channels.

The measured mobility of $\mu \approx 9.5 \times 10^5\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ places the sample in a regime that is clean enough to resolve Shubnikov–de Haas oscillations and the $\nu = 2$ Hall plateau, but not optimised for observation of higher filling-factor plateaux under the present experimental conditions. This classification is consistent with reported mobilities for GaAs/AlGaAs heterostructures in laboratory-scale and early research experiments [4].

The transport lifetime extracted from the mobility is also compatible with the magnetic-field scale at which quantum oscillations first appear, providing further internal consistency between the longitudinal transport and quantum oscillation analyses.

5.4 Robustness and Internal Consistency of the Analysis

A key strength of the present analysis is the internal consistency between transport parameters extracted from independent classical and quantum measurements. Carrier densities obtained from the low-field Hall slope, Shubnikov–de Haas periodicity, Landau fan analysis, and the $\nu = 2$ plateau position agree within approximately 0.3 %, providing strong validation of the magnetic-field calibration and filling-factor assignment.

The use of the $\nu = 2$ Hall plateau as an absolute reference for current calibration further enhances this internal consistency. Adopting the plateau-derived current yields mutually consistent carrier densities and resistivity conversions across all analysis methods, whereas reliance on instrument readback alone introduces systematic discrepancies.

Additionally, the agreement between up-sweep and down-sweep measurements across the $\nu = 2$ plateau indicates that the observed quantisation is stable and not sensitive to sweep history or measurement-induced heating.

5.5 Uncertainties and Experimental Limitations

The dominant source of uncertainty in the extracted mobility arises from the Hall-bar geometry factor W/L, rather than from electronic noise or fitting uncertainty. This systematic geometric uncertainty propagates directly into ρ_{xx} and mobility estimates but does not affect the carrier density extracted from Hall slopes or quantum oscillation periodicities.

A constant scaling offset of approximately 5 % was identified in one Hall-voltage channel due to an instrument configuration setting and corrected prior to parameter extraction. This correction does not alter qualitative transport trends and improves agreement between independent carrier-density estimates, indicating that the principal physical conclusions are robust against this instrument configuration setting.

A leak in the cryostat vacuum space prevented stable operation at the nominal base temperature of the system. To maintain thermal stability, an active temperature stabilisation was used, limiting the lowest achievable temperature to approximately 3 K. This temperature limitation reduces the resolution of higher filling-factor features and restricts the temperature-dependence analysis to qualitative trends but does not preclude observation of the defining $\nu = 2$ quantum Hall signatures.

5.6 Scope, Interpretation Limits, and Future Improvements

Only the $\nu = 2$ Hall plateau is clearly resolved in the present measurements, as expected given the combined effects of finite temperature, moderate disorder, and the accessible magnetic-field range. No metrological claim is made; although the Hall resistance is quantised to within tens of parts per million, the measurements represent a laboratory-scale demonstration of the integer quantum Hall effect rather than a precision resistance standard.

Future measurements could improve precision and extend the range of observable filling factors by achieving lower and more stable base temperatures, improving Hall-bar geometry characterisation, and enabling measurements over a wider temperature range. Such improvements would allow a full Lifshitz–Kosevich analysis of the Shubnikov–de Haas oscillations and a more detailed investigation of disorder effects.

6 Conclusion

In this project, magnetotransport measurements were performed on a GaAs/AlGaAs two-dimensional electron gas to investigate the integer quantum Hall effect under laboratory-scale cryogenic conditions. The aim was to demonstrate the defining transport signatures of the effect and to extract key electronic parameters using internally consistent classical and quantum analysis methods.

A clear Hall resistance plateau was observed at filling factor $\nu = 2$, centred at a magnetic field of $B_{\nu=2} = 3.23\,\mathrm{T}$, accompanied by strong suppression of the longitudinal resistance. Within the plateau window, the measured Hall resistance deviates from the ideal quantised value $h/2e^2$ by only tens of parts per million. Carrier densities extracted from four independent methods cluster around $n_s = 1.57 \times 10^{15}\,\mathrm{m}^{-2}$ with a maximum fractional spread of approximately $0.3\,\%$, demonstrating strong internal consistency.

From low-field longitudinal transport, a mobility of $\mu \approx 9.5 \times 10^5\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$ and a corresponding transport lifetime of $\tau_{\mathrm{tr}} \approx 3.5 \times 10^{-11}\,\mathrm{s}$ were obtained. The temperature dependence of the Shubnikov–de Haas oscillations is consistent with thermal broadening of Landau levels, while the invariance of the low-field Hall slope confirms that the carrier density remains constant over the temperature range studied.

Despite experimental constraints on temperature stability and geometric uncertainty, the essential signatures of the integer quantum Hall effect are robustly observed. This work demonstrates that laboratory-scale cryogenic apparatus is sufficient to access the quantum Hall regime and to perform a quantitatively consistent analysis of two-dimensional electron transport. Future improvements in cryogenic performance and device characterisation would enable higher precision and extended exploration of quantum Hall physics.

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Appendices

The appendices provide supporting diagnostics and methodological details that complement the main Results section. Detailed analysis scripts, exploratory plots, and intermediate checks are documented in the project diary and are not reproduced in full here.

A Data processing, corrections, and numerical stability

A.1 Conversion of raw voltage signals to resistance

During magnetotransport measurements, the raw quantities recorded for each field sweep were the longitudinal voltage V_{xx}, the transverse (Hall) voltage V_{xy}, the perpendicular magnetic field B, and the sample temperature T. These quantities were acquired simultaneously using lock-in detection and time-stamped to ensure direct correspondence between voltage, field, and temperature at each data point.

Measured voltages were converted to longitudinal and Hall resistances using the relation

$$R = \frac{V}{I}. \tag{11}$$

where I is the effective excitation current flowing through the Hall bar. The same operating current was used consistently for both R_{xx} and R_{xy} conversion within a given dataset, and all voltage-to-resistance conversions were performed prior to any further analysis steps.

The polarity conventions for V_{xx} and V_{xy} were defined by the wiring configuration shown in Fig. 4 and were held fixed across all measurements. The sign of R_{xy} reverses under magnetic-field reversal, while R_{xx} remains an even function of B, providing an internal consistency check on voltage-channel assignment and wiring polarity. No sign ambiguities or wiring-dependent offsets were observed in the converted resistance data.

The conversion from raw voltages to resistances is unambiguous; any remaining systematic uncertainties arise from calibration factors discussed elsewhere rather than from the conversion procedure itself.

A.2 Identification and correction of a Hall-voltage scaling offset

A constant multiplicative scaling offset of approximately 5 % was identified in one V_{xy} channel during verification of analysis consistency. The presence of this offset was indicated by a systematic mismatch between carrier-density values extracted from independent methods (low-field Hall slope, SdH periodicity/Landau fan, and the $\nu = 2$ plateau position), when all other analysis inputs and conventions were held fixed. Because the discrepancy was consistent across field sweeps, temperatures, and analysis routes, it was treated as a channel-wide scale factor rather than a field-dependent distortion.

The affected V_{xy} channel was corrected by applying a uniform multiplicative factor to the recorded voltage prior to conversion into R_{xy} and prior to all subsequent parameter extraction. After this correction, carrier-density estimates from independent methods became mutually consistent within their quoted uncertainties, while the qualitative magnetotransport signatures (Hall staircase, SdH oscillations, and the $\nu = 2$ plateau with suppression of R_{xx}) were unchanged. All results and figures reported in the main text are based on the corrected dataset.

A.3 Conditioning of resistivity–conductivity tensor inversion

The conductivity tensor components were obtained from the resistivity tensor via the inversion in Eq. (8). This transformation is applied uniformly to the full resistivity dataset following conversion from measured resistances using the Hall-bar geometry factor. The inversion enables direct comparison with theoretical descriptions of quantum Hall transport formulated in terms of conductivity tensors.

As the magnetic field approaches zero, the Hall resistivity ρ_{xy} also approaches zero, causing the denominator $\rho_{xx}^2 + \rho_{xy}^2$ to become small. In this limit, the tensor inversion becomes numerically sensitive to measurement noise and small uncertainties in both ρ_{xx} and ρ_{xy}, leading to amplified fluctuations in the computed conductivity components. This sensitivity is a mathematical consequence of the inversion procedure rather than a physical feature of the transport.

To avoid over-interpretation of numerically ill-conditioned values, the immediate low-field region near $B = 0$ was excluded from quantitative analysis of σ_{xx} and σ_{xy} and is shown as a masked region in the conductivity plots. Outside this region, the inversion is well conditioned and yields physically meaningful features, including minima in σ_{xx} coincident with plateaux in σ_{xy}. These features are therefore used for qualitative comparison with localisation-based interpretations of the integer quantum Hall effect, as discussed in Secs. 5.5 and 6.2.

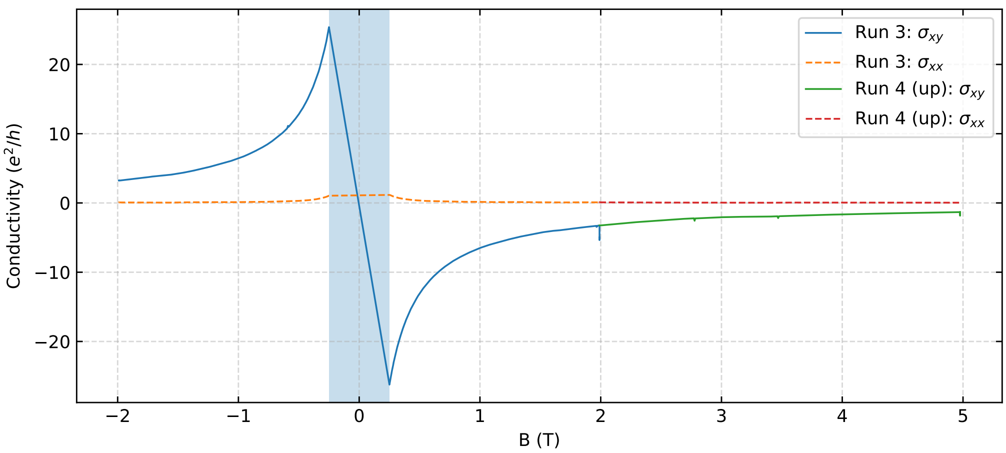

Figure 14: Conductivity tensor components $\sigma_{xx}(B)$ and $\sigma_{xy}(B)$ obtained by inversion of the resistivity tensor. The shaded region around $B = 0$ indicates the low-field interval excluded from quantitative interpretation due to numerical conditioning of the inversion. Outside this region, minima in σ_{xx} coincide with plateaux in σ_{xy}.

B Robustness and consistency checks

B.1 Definition of the $\nu = 2$ Hall plateau window

The magnetic-field window associated with the $\nu = 2$ Hall plateau was defined using a flat-region criterion applied to the measured Hall resistance $R_{xy}(B)$. Specifically, the plateau window was identified as the contiguous field interval over which the local slope

$\mathrm{d}R_{xy}/\mathrm{d}B$ is consistent with zero within the experimental noise level, indicating that R_{xy} remains approximately constant with magnetic field. The resulting plateau bounds define the window used for extraction of the plateau-averaged Hall resistance and the central plateau field reported in Sec. 5.3.

Modest variations of the plateau window produce changes in the plateau-averaged R_{xy} that are small compared to the quoted deviation from $h/2e^2$, confirming robustness against the precise window definition.

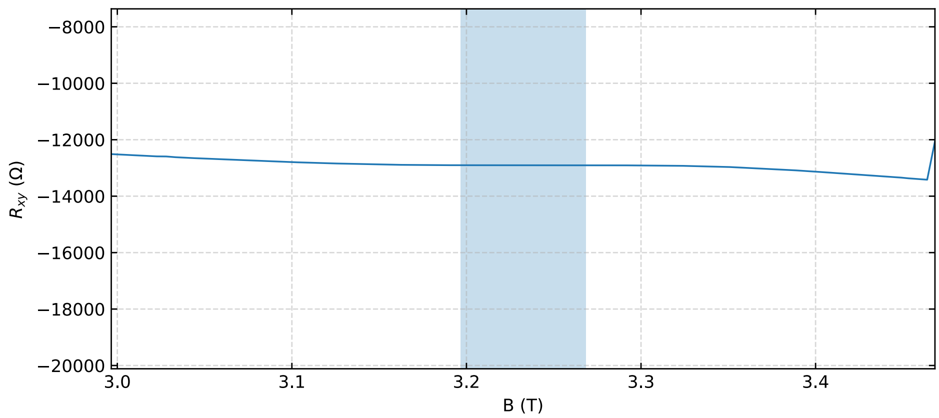

Figure 15: Zoomed view of the Hall resistance $R_{xy}(B)$ in the vicinity of the $\nu = 2$ plateau. The shaded region indicates the magnetic-field window over which the plateau criterion is satisfied and from which plateau-averaged quantities are extracted.

B.2 Up-sweep and down-sweep reproducibility

To assess the reproducibility of the Hall resistance quantisation and to test for possible sweep-direction-dependent effects, the $\nu = 2$ plateau was examined separately for magnetic-field up-sweeps and down-sweeps. Comparing sweep directions provides a check for hysteresis, slow thermal equilibration, or measurement-induced heating that could otherwise bias the plateau-averaged Hall resistance.

For each sweep direction, the deviation of the measured Hall resistance from the ideal quantised value $h/2e^2$ was computed across the plateau window defined in Appendix B.1. The up-sweep and down-sweep traces exhibit quantitatively similar deviations, with no systematic offset beyond experimental uncertainty.

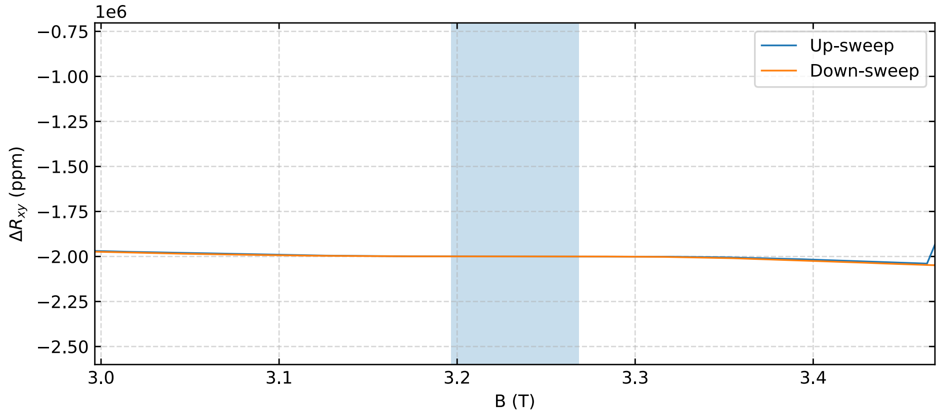

Figure 16: Deviation of the measured Hall resistance from the ideal quantised value $h/2e^2$ within the $\nu = 2$ plateau window for magnetic-field up-sweeps and down-sweeps. The close agreement between sweeps indicates negligible hysteresis or sweep-direction-dependent effects.

B.3 Consistency of operating-current calibration

The excitation current inferred from the lock-in readback and that derived from the $\nu = 2$ Hall plateau differ by a constant factor but remain stable across magnetic-field sweeps and temperatures.

C Carrier density extraction diagnostics

C.1 Low-field Hall slope determination

The carrier density was extracted from a linear fit to the low-field Hall response $R_{xy}(B)$ in a magnetic-field regime where classical transport dominates and quantum oscillations are absent.

Within this interval, the Hall resistance was modelled using the classical relation from Eq. (5) and the carrier density n_s was obtained from the slope of a least-squares linear fit. Uncertainties were estimated from the standard error of the fitted slope and propagated directly to the extracted carrier density.

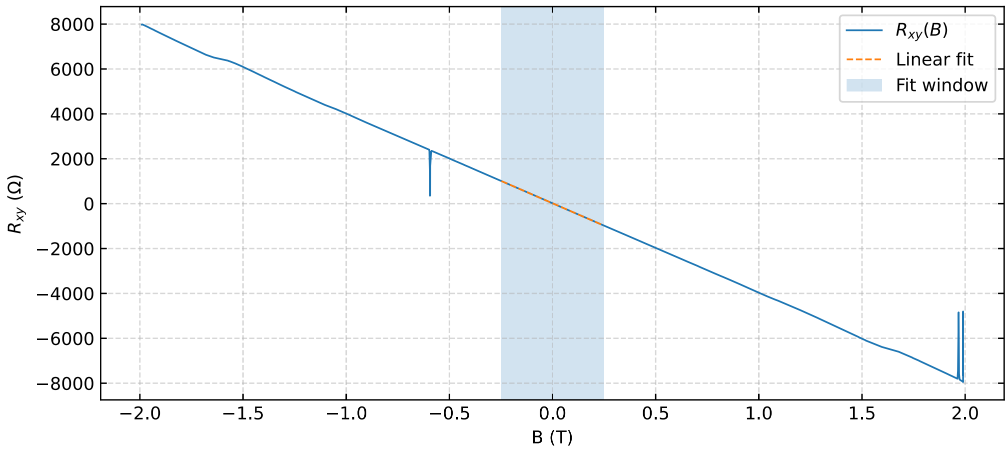

Figure 17: Low-field Hall resistance $R_{xy}(B)$ with linear least-squares fit used to extract the carrier density from the classical Hall slope. The shaded region indicates the magnetic-field interval included in the fit; higher-field data exhibiting quantum deviations were excluded.

C.2 Shubnikov–de Haas peak identification and oscillation periodicity

Quantum-mechanical carrier density was extracted from the periodicity of Shubnikov–de Haas (SdH) oscillations by identifying extrema in the longitudinal resistance and analysing their spacing in inverse magnetic field. The analysis was performed on R_{xx} data plotted as a function of $1/B$, where SdH oscillations are expected to be periodic for a single two-dimensional carrier population. Only data in the magnetic-field range where oscillations are clearly resolved above the noise floor were included.

Oscillation extrema were identified using an automated peak-finding procedure, with manual verification confirming that peak positions are robust against reasonable variations in smoothing parameters.

The inverse-field spacing $\Delta(1/B)$ between successive extrema is approximately constant over the analysed field range, indicating transport governed by a single carrier density. The extracted periodicity is consistent with values obtained from the low-field Hall slope, Landau fan analysis, and the $\nu = 2$ plateau position reported in Sec. 5.2.

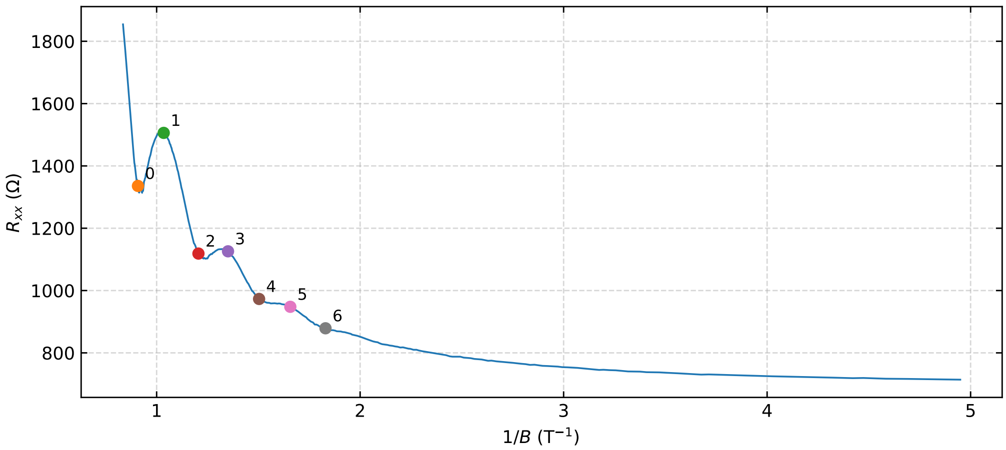

Figure 18: Longitudinal resistance R_{xx} plotted as a function of inverse magnetic field $1/B$, showing indexed Shubnikov–de Haas oscillation extrema used to determine the oscillation periodicity $\Delta(1/B)$. The near-uniform spacing of peaks supports extraction of a single carrier density.

C.3 Landau fan analysis

An independent quantum-mechanical determination of the carrier density was obtained using a Landau fan analysis, in which the indices of Shubnikov–de Haas oscillation extrema are plotted as a function of inverse magnetic field. Extrema in R_{xx} were assigned integer indices N corresponding to successive oscillations, following the standard convention that adjacent extrema differ by one in index. The analysis was restricted to the magnetic-field range where oscillations are well resolved and indexing is unambiguous.

The indexed extrema were plotted as a function of $1/B$, and a linear least-squares fit was applied to the resulting fan diagram. The slope of this relation provides an independent determination of the carrier density.

The second derivative of R_{xx} with respect to magnetic field was used as a visual diagnostic to aid identification of oscillation extrema without altering their positions.

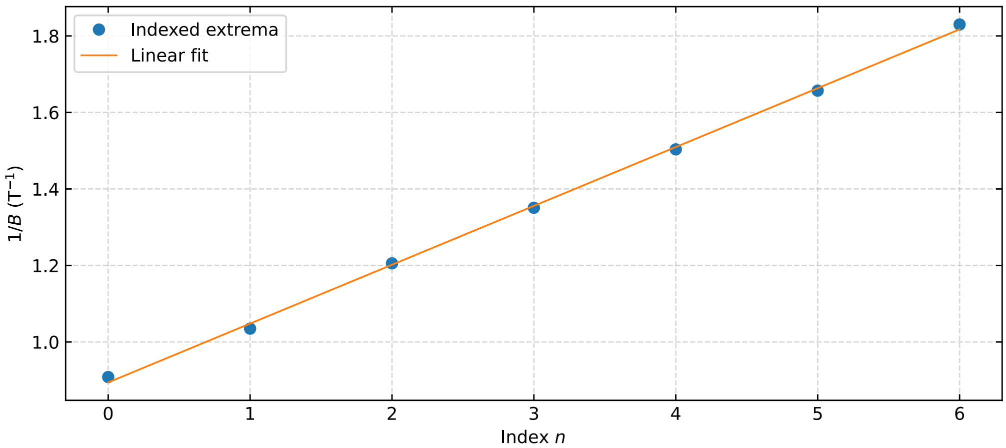

Figure 19: Landau fan diagram showing indexed Shubnikov–de Haas oscillation extrema plotted as a function of inverse magnetic field $1/B$. The linear fit to the data provides an independent determination of the carrier density from the slope of the fan.

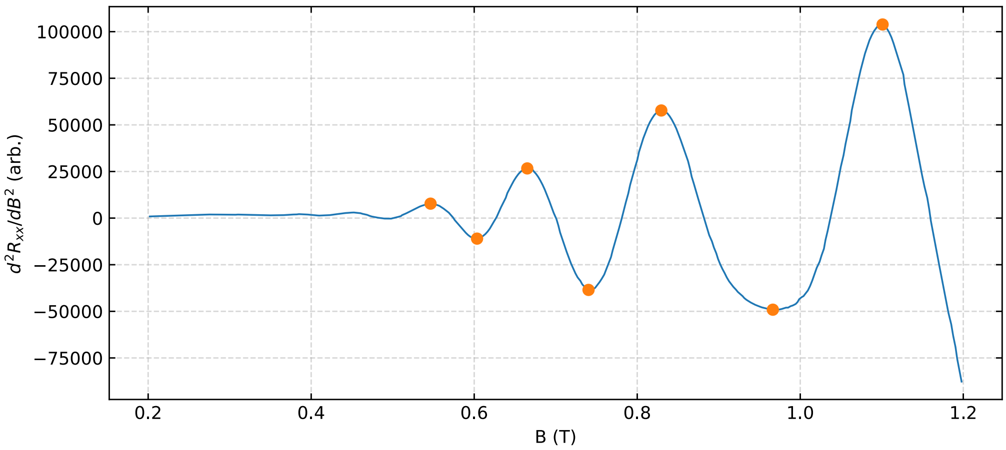

Figure 20: Auxiliary diagnostic plot showing the second derivative of the longitudinal resistance with respect to magnetic field. The enhanced contrast aids identification of Shubnikov–de Haas extrema used in the Landau fan construction.

D Geometry factor and uncertainty propagation

D.1 Hall-bar geometry measurement

The conversion of measured longitudinal resistance to resistivity requires knowledge of the Hall-bar geometry through the ratio W/L, where W is the width of the conducting channel and L is the separation between the longitudinal voltage probes. Both quantities were measured directly from the sample using optical inspection and calibrated scale references. The resulting geometry factor was treated as constant across all measurements.

These effects introduce uncertainties that are comparable in magnitude across measurements and are not reduced by averaging.

Because the same geometry factor is applied uniformly to all longitudinal transport data, the associated uncertainty is systematic rather than statistical. This uncertainty propagates directly into quantities derived from ρ_{xx}, most notably the mobility, while leaving quantities extracted from transverse transport unaffected. It therefore sets the dominant precision limit for longitudinal transport parameters reported in Sec. 5.6.

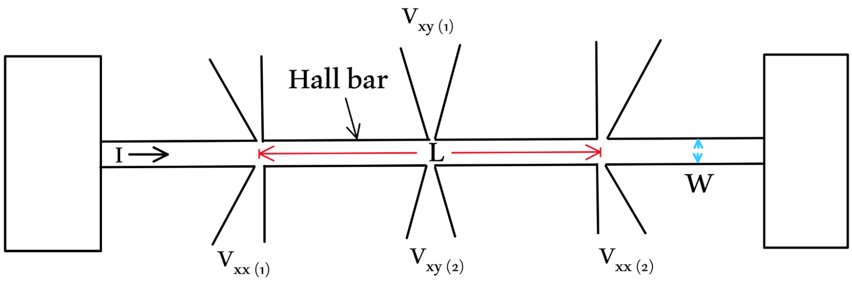

Figure 21: Annotated schematic of the Hall-bar geometry indicating the channel width W and longitudinal probe separation L used to determine the geometry factor W/L for conversion of resistance to resistivity.

D.2 Propagation of uncertainty in mobility determination

The carrier mobility μ was determined from longitudinal transport using the relation from Eq. (9) where ρ_{xx} is obtained from the measured longitudinal resistance through the Hall-bar geometry factor W/L. Uncertainty in μ therefore arises from contributions associated with the geometry factor, the measured resistance, and the independently determined carrier density.

The dominant contribution to the uncertainty in ρ_{xx}, and hence in μ, originates from the systematic uncertainty in the Hall-bar geometry factor W/L described in Appendix D.1. In comparison, voltage noise in the longitudinal resistance measurements produces only small point-to-point fluctuations that average out over field sweeps and do not contribute significantly to the final uncertainty. Propagation of uncertainties confirms that electronic noise is subdominant relative to geometric effects in the determination of the mobility.

The carrier density n_s enters the mobility expression as an independently determined parameter extracted from transverse transport measurements. As discussed in Sec. 5.2 and Appendix C, n_s is obtained from Hall and quantum oscillation analyses that do not depend on the Hall-bar geometry factor. Consequently, uncertainty in W/L does not propagate into the carrier density and affects only quantities derived from longitudinal transport.